

# Reliability-Aware Adaptive Cross-Modal Fusion for Robust Multimodal Sentiment Analysis

\*1<sup>st</sup> Xingao ChenSchool of Computer Science and  
TechnologyWuhan University of Science and  
Technology

Wuhan, China

1176032422@qq.com

Qintong Wang

School of Computer Science and  
TechnologyWuhan University of Science and  
Technology

Wuhan, China

202413201069@wust.edu.cn

Ruoxin Zhan

School of Computer Science and  
TechnologyWuhan University of Science and  
Technology

Wuhan, China

paracosm@wust.edu.cn

Baixin Chen

School of Computer Science and  
TechnologyWuhan University of Science and  
Technology

Wuhan, China

2844047610@qq.com

Yuxuan Zhu

School of Computer Science and  
TechnologyWuhan University of Science and  
Technology

Wuhan, China

1447578110@qq.com

**Abstract**—Existing multimodal fusion methods usually assume that the contribution of each modality is symmetrical, which often leads to redundant interactions and sensitivity to environmental noise. In this paper, we propose a Reliability-Aware Adaptive Cross-Modal Fusion Network (RAF-Net) to address these challenges. Instead of treating all signals equally, RAF-Net processes language and acoustic streams as dense semantic carriers, while utilizing visual information as a static reliability anchor to stabilize predictions. To support this robust framework, we introduce Sparse Interaction Regularization to prune redundant cross-modal correlations. Crucially, a Reliability-Aware Decision Module is deployed to dynamically calibrate the contribution of each modality based on context confidence. Experiments on the CMU-MOSI and CMU-MOSEI datasets demonstrate that RAF-Net outperforms state-of-the-art baselines, verifying its effectiveness in identifying significant emotional clues under noisy conditions.

**Keywords**—Multimodal Sentiment Analysis, Multimodal Fusion, Contrastive Learning, Reliability-Aware Decision

## I. INTRODUCTION

Sentiment analysis is a core paradigm in natural language processing (NLP), aimed at interpreting human subjective emotions and opinions from unstructured data. With the proliferation of multimedia content [1], emotional expression has evolved beyond text to encompass the synergy of acoustic and visual signals [2]. Consequently, Multimodal Sentiment Analysis (MSA) has emerged as a crucial field, leveraging the complementarity between heterogeneous modalities to achieve a more comprehensive understanding than unimodal methods.

Despite significant progress, most existing fusion paradigms—from early tensor-based methods [3], [4] to recent Transformer-based cross-attention architectures [5]—implicitly assume that all modalities contribute equally and should be processed via symmetrical interaction mechanisms. However, in real-world scenarios, this assumption often fails due to the prevalence of intra-modal redundancy and contextual noise [6]. Applying uniformly dense cross-modal interactions to such heterogeneous signals inevitably introduces computational overhead and propagates background noise, potentially overshadowing core emotional content.

To address these challenges, we propose a robust framework titled Reliability-Aware Adaptive Cross-Modal Fusion Network (RAF-Net). Notably, we identify text and acoustic modalities serve as dense semantic carriers, and visual modalities as static reliability anchors. Based on this, RAF-Net incorporates a Sparse Interaction Regularization mechanism to actively prune irrelevant cross-modal connections, preventing noise propagation at the feature level. Furthermore, to handle the varying quality of input signals, we design a Reliability-Aware Decision Module. This module acts as a dynamic gatekeeper, estimating the confidence of each modality in real-time and adaptively recalibrating their fusion weights. The main contributions of this work are summarized as follows:

- We identify the limitations of symmetric fusion in noisy environments and propose RAF-Net, a unified framework that enhances robustness through a reliability-aware adaptive mechanism.
- We introduce Sparse Interaction Regularization to suppress redundant cross-modal correlations, and

effectively utilize visual information as a global reliability anchor to stabilize predictions.

- We design a Reliability-Aware Decision Module that dynamically calibrates the contribution of each modality based on context confidence, ensuring precise decision-making even under uncertain conditions.
- Extensive experiments on the benchmark CMU-MOSI and CMU-MOSEI datasets demonstrate that RAF-Net outperforms state-of-the-art baselines, particularly in noise-intensive scenarios.

## II. RELATED WORKS

### A. Multimodal Sentiment Analysis

Multimodal sentiment analysis aims to integrate emotional clues from heterogeneous data sources to achieve stable emotional understanding [7]. Compared with the single-modal method, MSA benefits from cross-modal complementarity, which helps to eliminate modal specificity ambiguity. Early methods usually rely on feature-level concatenation or decision-level fusion [8], but these methods are difficult to model complex intramodal dynamics and high-order cross-modal interactions. Subsequent studies introduced deep interactive modeling, including tensor-based fusion [4], graph neural networks for unaligned sequence modeling [9], and other theoretical frameworks.

With the success of the Transformer model, the fusion method based on the attention mechanism occupies a dominant position in the field of multimodal sequence analysis (MSA). Multimodal Transformer (MulT) [10] took the lead in realizing the cross-modal attention mechanism and achieving implicit alignment between unaligned sequences. On this basis, MAG-BERT [11] and other methods introduce gate control mechanisms and use non-verbal clues to adjust text characterization; while the latest models such as MML [12] adopt multiple loss optimization to improve robustness.

### B. Robustness and Adaptive Fusion

Multimodal sentiment analysis needs to achieve effective semantic alignment between heterogeneous modalities while maintaining robustness to widespread noise and redundancy. Early research addressed alignment through shared subspace learning and feature decoupling, such as MISA [13], which separates modality-invariant and specific features, and Self-MM [14], which introduces self-supervised auxiliary targets to guide unimodal encoders. In recent years, contrastive learning has become a mainstream strategy by maximizing cross-modal mutual information to enforce stricter semantic consistency, as demonstrated by UniMSE [15].

However, real-world data is plagued by intra-modal redundancy and specific uncertainty. While regularization methods like multimodal Dropout and sparse attention alleviate overfitting, they often rely on static or heuristic constraints. Dynamic fusion models, such as MM-DFN [16], introduce gating or uncertainty estimation; however, operations based on unrefined features usually lead to noise propagation and unreliable fusion. In general, most existing methods implicitly assume that all input modalities are equally reliable, lacking

dynamic mechanisms to down-weight noisy signals based on context.

## III. METHODOLOGY

### A. Task Definition

Formally, we formulate the multimodal sentiment analysis task as a regression problem, aiming to utilize heterogeneous inputs—Language (T), Acoustics (A), and Visual (V)—to predict the sentiment intensity of a specific discourse  $l$ . Consider a video sequence denoted as  $S = \{u_1, u_2, \dots, u_L\}$ . For a given target discourse  $u_i$ , the model handles different feature sets  $\{X_i^t, X_i^a, X_i^v\}$ , which correspond to text marks, acoustic features and visual frames respectively.

Unlike standard point-wise prediction methods, a prominent feature of our approach is the introduction of a backtracking context mechanism. Instead of processing utterances in isolation, we construct a context window of size  $k$  for each target  $u_i$ . Consequently, the input comprises not only direct features but also a historical context set  $C_i = \{u_{i-k}, \dots, u_{i-1}\}$ , which aids in resolving local semantic ambiguities. The primary objective of RAF-Net is to approximate the mapping function  $\mathcal{F}$  parameterized by  $\Theta$ , defined as:

$$\hat{y}_i = \mathcal{F}(X_i^t, X_i^a, X_i^v, C_i; \Theta) \quad (1)$$

where  $\hat{y}_i \in \mathbb{R}$  represents the estimated sentiment intensity. The training goal is to minimize the regression error between the prediction  $y_i$  and the ground truth label.

### B. Overall Architecture

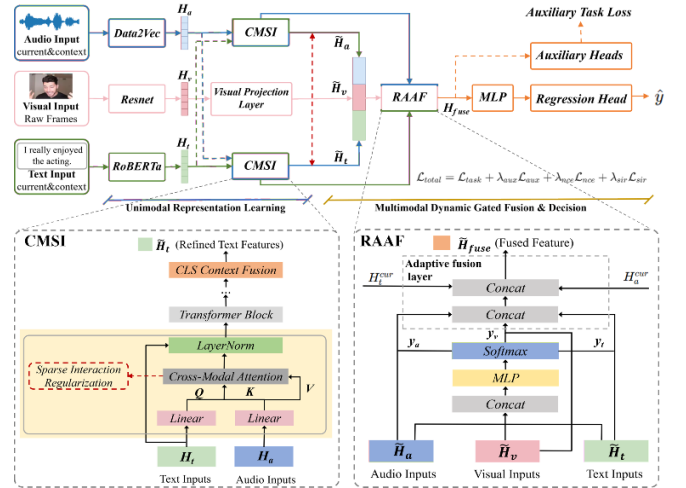


Fig. 1. The overall architecture of the proposed RAF-Net

The overall structure of RAF-Net is shown in Fig. 1. For text and audio streams, the architecture adopts a strict refinement process in the cross-modal sparse interaction (CMT) module and uses a sliding context window to process it. Here, we realize sparse interaction regularization and contrast alignment to actively eliminate noise and ensure semantic consistency. At the same time, the visual flow bypasses this heavy interaction logic and extracts the static global reliability anchor point through a simplified projection layer. Finally, the signal gathers in the reliability perception adaptive fusion (RAAF) module. In this

module, the visual anchor point dynamically recalibrates the multimodal reliability, and the representation of the final fusion is supervised by the multitasking assistant head to ensure semantic consistency, and then the final regression is carried out

### C. Feature extract module

The feature extraction protocol adopts a hierarchical structure to process heterogeneous signals. We deploy high-capacity encoders for text and audio to retain fine-grained details, while adopting a streamlined approach for visual anchors.

Text and Audio Linguistic representations are harvested using the pre-trained RoBERTa [17]. For every instance  $i$ , the model processes parallel input streams: the immediate utterance  $X_{cur}^t$  and the historical context  $X_{ctx}^t$ . We extract sentence-level embeddings directly via the [CLS] token:

$$H_{cur}^t = \text{RoBERTa}(X_{cur}^t)_{[\text{CLS}]} \quad (2)$$

$$H_{ctx}^t = \text{RoBERTa}(X_{ctx}^t)_{[\text{CLS}]} \quad (3)$$

where  $H_{cur}^t, H_{ctx}^t \in \mathbb{R}^{d_t}$  are the extracted dense semantic features.

To capture prosodic cues, we employ Data2Vec [18]. Given raw waveforms  $X_{cur}^a$  and  $X_{ctx}^a$ , the encoder outputs frame-level hidden states. To filter out padding noise, we apply Masked Mean-Pooling determined by the attention mask  $M$ :

$$H^a = \frac{1}{\sum_{j=1}^L M_j} \sum_{j=1}^L M_j \cdot \text{Data2Vec}(X^a)_j \quad (4)$$

Applying this operation to both streams yields the robust acoustic representations  $H_{cur}^a$  and  $H_{ctx}^a$ .

Given that visual cues in talking-head videos are often spatially redundant, we treat the visual modality as a stationary reliability anchor rather than performing expensive sequence modeling. For a video sequence of  $K$  frames  $I = \{I_1, I_2, \dots, I_K\}$ , we utilize ResNet-50 [19] with temporal average pooling to obtain a unified representation:

$$H_{raw}^v = \frac{1}{K} \sum_{k=1}^K \text{ResNet50}(I_k) \quad (5)$$

To align this feature with the dense semantic space, we pass it through a projection layer consisting of Layer Normalization and a linear transformation:

$$\tilde{H}_v = W_v \cdot \text{LayerNorm}(H_{raw}^v) + b_v \quad (6)$$

The resulting  $\tilde{H}_v$  serves as the stationary reliability anchor in the RAAF module.

### D. Cross-Modal Sparse Interaction

To coordinate language semantics and acoustic semantics while suppressing noise, this module goes beyond simple splicing. It adopts a dual-stream scheme to jointly refine the representation of the current discourse and its historical context.

Let the input features be  $H^t \in \{H_{cur}^t, H_{ctx}^t\}$  and  $H^a \in \{H_{cur}^a, H_{ctx}^a\}$ . We employ a stack of  $L$  encoder layers to perform

bi-directional interaction. For any layer, text features serve as the *Query* to retrieve complementary details from Audio Key/Value pairs, and vice versa. This effectively aligns prosodic intensity with linguistic semantics:

$$\hat{H}_{t \rightarrow a}^{(l)} = \text{MHA}(Q = H_t^{(l-1)}, K = H_a^{(l-1)}, V = H_a^{(l-1)}) \quad (7)$$

$$\hat{H}_{a \rightarrow t}^{(l)} = \text{MHA}(Q = H_a^{(l-1)}, K = H_t^{(l-1)}, V = H_t^{(l-1)}) \quad (8)$$

Following this exchange, a standard Self-Attention layer and a Feed-Forward Network (FFN) integrate the interactive features for each stream:

$$H_m^{(l)} = \text{FFN}(\text{SelfAttn}(\hat{H}_{m \rightarrow \cdot}^{(l)})), m \in \{t, a\} \quad (9)$$

The cross-attention diagram is supervised by sparse interaction regularization. This enforces adaptive, data-driven sparseness, enabling the model to focus on semantically meaningful alignment without applying rigid structural constraints

Finally, the refined representations from the current utterance and its context are aggregated via mean pooling to obtain the final features  $\tilde{H}_t, \tilde{H}_a$ :

$$\tilde{H}_t = \text{Mean}(H_{cur}^t, H_{ctx}^t), \tilde{H}_a = \text{Mean}(H_{cur}^a, H_{ctx}^a) \quad (10)$$

### E. Reliability-Aware Adaptive Fusion

To solve the problem of reliability deviation, the module acts as a dynamic arbitrator. It uses static visual anchors as a stable reference to recalibrate the contribution of each data stream.

First, we construct a unified multimodal descriptor  $G_{in}$  by concatenating the refined semantic features with the visual anchor:

$$G_{in} = \text{Concat}(\tilde{H}_t, \tilde{H}_a, \tilde{H}_v) \quad (11)$$

To quantify modal confidence, we design a lightweight gating network: taking a global descriptor as input, mapping it through a multilayer perceptron (MLP) and normalizing it with Softmax to generate the modal weights.:

$$[y_t, y_a, y_v] = \text{Softmax}(\text{MLP}_{gate}(G_{in})) \quad (12)$$

where  $y_m$  represents the importance weight of modality  $m$ .

The final fusion integrates these weighted features. To prevent the gating mechanism from filtering out too much intrinsic information, we introduce a Contextual Residual Connection by concatenating the original raw features back into the fusion:

$$H_{core} = \text{Concat}(y_t \cdot \tilde{H}_t, y_a \cdot \tilde{H}_a, y_v \cdot \tilde{H}_v) \quad (13)$$

$$H_{fuse} = \text{Concat}(H_{core}, H_{cur}^t, H_{cur}^a) \quad (14)$$

Finally,  $H_{fuse}$  is projected through a regression head to generate the sentiment prediction  $\hat{y}$

### F. Prediction and Optimization Objectives

The fused representation  $H_{fuse}$  from the RAAF module is projected into the target sentiment space via a prediction head. This head consists of a 3-layer Multilayer Perceptron with ReLU activation and Dropout to ensure non-linear mapping capabilities:

$$\hat{y} = W_3 \cdot \sigma(W_2 \cdot \sigma(W_1 \cdot H_{fuse} + b_1) + b_2) + b_3 \quad (15)$$

where  $\sigma$  denotes the ReLU activation, and  $\hat{y} \in \mathbb{R}$  is the predicted sentiment intensity score.

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda_{sir}\mathcal{L}_{sir} + \lambda_{nce}\mathcal{L}_{nce} + \lambda_{aux}\mathcal{L}_{aux} \quad (16)$$

To train RAF-Net effectively, we employ a composite loss function balancing regression accuracy with feature refinement. The total objective  $\mathcal{L}_{total}$  is formulated as:

$$\mathcal{L}_{task} = \sum_{m \in \{M, T, A\}} \|\hat{y}_m - y\|_1 \quad (17)$$

To suppress noise in the CMSI module, we minimize the entropy of the attention maps  $\mathcal{A}$ . This forces the model to make confident decisions by focusing on semantically relevant frames rather than attending uniformly to noisy backgrounds:

$$\mathcal{L}_{sir} = \frac{1}{N_h} \sum_{h=1}^{N_h} \left( - \sum_j \mathcal{A}_{h,j} \log(\mathcal{A}_{h,j} + \epsilon) \right) \quad (18)$$

We employ InfoNCE loss to bridge the semantic gap between dual-stream encoders. This maximizes the mutual information between refined text  $\tilde{H}_t$  and audio  $\tilde{H}_a$  representations of the same sample, ensuring semantic consistency:

$$\mathcal{L}_{nce} = -\log \frac{\exp(\text{sim}(\tilde{H}_t, \tilde{H}_a)/\tau)}{\sum_{k=1}^B \exp(\text{sim}(\tilde{H}_t, \tilde{H}_{a,k})/\tau)} \quad (19)$$

To stabilize training, we add auxiliary classification heads (binary and 7-class) to the fusion features. Optimized via cross-entropy loss, these provide coarse-grained supervision to guide the fine-grained regression task.

## IV. EXPERIMENTS

### A. Datasets

To strictly evaluate the noise resistance and generalization of RAF-Net, we conduct experiments on two standard benchmarks: CMU-MOSI [20] and CMU-MOSEI [21]. A key distinction of our setup is the direct processing of original multimodal signals rather than pre-aligned features, enabling a rigorous end-to-end evaluation on unaligned real-world data. Detailed statistics are provided in Table 1.

TABLE I. STATISTICAL BREAKDOWN OF THE CMU-MOSI AND CMU-MOSEI DATASETS

| Dataset | Train  | Valid | Test | All    |
|---------|--------|-------|------|--------|
| MOSI    | 1284   | 229   | 686  | 2199   |
| MOSEI   | 16,326 | 1871  | 4659 | 22,856 |

### B. Evaluation metrics

To ensure strict comparability with existing literature, our framework adopts a dual-index strategy covering emotional intensity regression and polarity classification. For regression, we employ Mean Absolute Error (MAE) and Pearson Correlation (Corr) to evaluate predictive fidelity and trajectory synchronization. For classification, we report 7-class accuracy (Acc-7) for fine-grained evaluation, along with Binary Accuracy (Acc-2) and F1 score to test polarity robustness. Consistent with established benchmarks, we adopt two binary mapping protocols: Has0 (Non-negative, treating neutral as positive where  $y \geq 0$ ) and Non0 (Strict, excluding vague neutral samples to focus on  $y < 0$  vs  $y > 0$ ). Unless stated otherwise, binary metrics are presented in the format "Has0 / Non0".

### C. Implement details

Our framework is implemented in PyTorch on an NVIDIA GeForce RTX 3090 GPU. Hyperparameters are tailored to the noise characteristics of each dataset. For CMU-MOSI, convergence is stable at a learning rate of  $5e-6$  with 20 epochs. Conversely, the noise-intensive CMU-MOSEI requires a more aggressive learning rate of  $8e-6$  (with 10 epochs) to escape local minima. Common settings across both datasets include a batch size of 8, dropout rate of 0.25, and early stopping patience of 8. Regarding architecture, we set the text context window to 2, audio context window to 1, and stack 5 cross-modal layers. The loss weights are strictly configured as  $\lambda_{sir} = 0.2$ ,  $\lambda_{nce} = 0.2$  and  $\lambda_{aux} = 0.2$ . To ensure reproducibility, all reported metrics represent the average of five independent runs using different random seeds.

### D. Baselines

To evaluate RAF-Net, we compare it against representative baselines across three categories:

- Tensor and Attention Fusion: TFN [3] captures multi-order dynamics via tensor outer products, while LMF [4] utilizes low-rank decomposition for efficiency. MulT [10] employs cross-modal attention to align unaligned sequences.
- Disentanglement and Gating: MISA [13] factorizes representations into invariant and specific subspaces to handle heterogeneity. MAG-BERT [11] integrates non-verbal cues into pre-trained language models via multimodal adaptation gates.
- Advanced Multi-Task and Contrastive Frameworks: Self-MM [14] leverages self-supervised multi-task learning for unimodal guidance. HyCon [22] proposes a unified learning framework. Focusing on noise suppression, ConFEDE [23] decomposes features to remove redundancy, while MMML [12] employs multi-loss optimization to enhance robustness against data anomalies.

### E. Results Analysis

The empirical data in Table 2 confirms our robustness hypothesis. RAF-Net outperforms baselines like MMML, achieving prominent Pearson Correlations of 0.843 on MOSI and 0.804 on MOSEI. Robustness is particularly evident on the noise-intensive CMU-MOSEI dataset. While traditional models

struggle with acoustic clutter, RAF-Net remains stable by yielding an MAE of 0.518 and a Non0 F1 score of 87.39%. We attribute this resilience to our reliability-aware design: by treating vision as a global reliability anchor and utilizing Sparse Interaction Regularization, the model effectively prevents noise propagation. Regarding binary polarity, RAF-Net sets new benchmarks. Specifically, on the MOSI dataset, it reaches F1 scores of 86.45% under the Has0 setting and 88.16% under the

Non0 setting. Similarly, on the stricter MOSEI Non0 protocol, it achieves a score of 87.39%. This decision boundary is driven by the CMSI module, where contrastive alignment minimizes critical polarity flip errors. Notably, the slight trade-off in fine-grained Acc-7 is deliberate. We filter out subtle yet noisy clues via reliability gating, prioritizing correlation consistency over sensitivity to granular labels. This strategy ensures superior generalization in unconstrained real-world scenarios.

TABLE II. COMPARISON OF THE PERFORMANCE OF RAF-NET WITH THE STATE-OF-THE-ART BASELINE ON THE CMU-MOSI AND CMU-MOSEI DATASET. IN ACC-2 AND F1, THE LEFT OF THE “/” IS CALCULATED AS “NEGATIVE/NON-NEGATIVE” AND THE RIGHT IS CALCULATED AS “NEGATIVE/POSITIVE”. NOTE: <sup>a</sup>: RESULTS FROM [15]; <sup>b</sup>: RESULTS FROM [23]; <sup>c</sup>: RESULTS FROM [12]

| <i>Model</i>               | <i>CMU-MOSI</i> |       |       |             |             | <i>CMU-MOSEI</i> |       |       |             |             |
|----------------------------|-----------------|-------|-------|-------------|-------------|------------------|-------|-------|-------------|-------------|
|                            | MAE             | Corr  | ACC-7 | ACC-2       | F1          | MAE              | Corr  | ACC-7 | ACC-2       | F1          |
| TFN <sup>a</sup> [3]       | 0.901           | 0.698 | 34.9  | - /80.80    | -/80.70     | 0.593            | 0.677 | 50.2  | -/82.50     | -/82.10     |
| LMF <sup>a</sup> [4]       | 0.917           | 0.695 | 33.2  | -/82.50     | -/82.40     | 0.623            | 0.700 | 48.0  | -/82.00     | -/82.10     |
| MuT <sup>a</sup> [10]      | 0.861           | 0.711 | -     | 81.50/84.10 | 80.60/83.90 | 0.580            | 0.713 | -     | -/82.50     | -/82.30     |
| MAG-BERT <sup>a</sup> [11] | 0.712           | 0.796 | -     | 84.20/86.10 | 84.10/86.00 | -                | -     | -     | 84.70/-     | 84.50/-     |
| MISA <sup>a</sup> [13]     | 0.804           | 0.764 | -     | 80.79/82.10 | 80.77/82.03 | 0.568            | 0.717 | -     | 82.59/84.23 | 82.67/83.97 |
| HyCon <sup>a</sup> [22]    | 0.713           | 0.790 | 46.60 | -/85.20     | -/85.10     | 0.601            | 0.776 | 52.80 | -/85.40     | -/85.60     |
| Self-MM <sup>b</sup> [14]  | 0.708           | 0.796 | 46.7  | 83.44/85.46 | 83.36/85.43 | 0.531            | 0.765 | 53.9  | 83.76/85.15 | 82.17/85.83 |
| ConFEDE <sup>b</sup> [23]  | 0.742           | 0.784 | 42.3  | 84.17/85.52 | 84.13/85.52 | 0.522            | 0.780 | 54.9  | 81.65/85.82 | 82.17/85.83 |
| MMML <sup>c</sup> [12]     | 0.643           | 0.838 | 48.3  | 85.91/88.16 | 85.85/88.15 | 0.517            | 0.791 | 55.0  | 86.32/86.73 | 86.23/86.49 |
| <b>RAF-Net</b>             | 0.698           | 0.843 | 45.4  | 86.44/88.11 | 86.45/88.16 | 0.518            | 0.804 | 53.8  | 83.39/87.42 | 83.80/87.39 |

#### F. Ablation and Robustness Analysis

a) *Impact of Key Components:* Table 3 shows that removing the RAAF module causes the most severe degradation (e.g., F1 drops from 87.39% to 72.45% on CMU-MOSEI), proving visual anchors are crucial to prevent spurious correlations in noisy environments. Furthermore, removing cross-modal interaction sharply reduces performance, highlighting its necessity in filtering intra-modal redundancy. Dropping only the contrastive constraint yields a moderate decline, indicating InfoNCE loss acts as a vital "compass" for strict semantic consistency during CMSI alignment. Finally, the slight drop in w/o  $\lambda_{aux}$  confirms auxiliary heads effectively prevent overfitting to label noise..

TABLE III. ABLATION STUDY OF RAF-NET ON THE CMU-MOSI AND CMU-MOSEI DATASET

| <i>Model</i>        | <i>CMU-MOSI</i> |              | <i>CMU-MOSEI</i> |
|---------------------|-----------------|--------------|------------------|
|                     | <i>ACC-7</i>    | <i>ACC-2</i> | <i>F1</i>        |
| <b>RAF-Net</b>      | 45.4            | 86.44/88.11  | 83.80/87.39      |
| w/o CMSI            | 40.5            | 81.60/83.45  | 78.75/82.40      |
| w/o RAAF            | 38.2            | 78.30/80.15  | 68.99/72.45      |
| w/o $\lambda_{nce}$ | 42.1            | 83.15/85.20  | 80.45/84.20      |
| w/o $\lambda_{aux}$ | 43.6            | 84.8 /86.50  | 81.90/85.82      |

b) *Robustness under Visual Perturbations:* To verify the role of visual mode as a static reliability anchor, we conducted

a disturbance experiment on the CMU-MOSEI dataset. As shown in Table 4. The results show that after replacing the visual input with chronologically disturbed frames, the performance of the model is comparable to that of the baseline model, indicating that the model relies on the global semantic context rather than the chronological order. On the contrary, the introduction of Gaussian noise will lead to a significant increase in average absolute error (MAE), which confirms that although the time dependence is small, the clarity of visual information is crucial for accurate emotional prediction.

TABLE IV. QUANTITATIVE BENCHMARKING OF RAF-NET'S ABILITY TO RESIST VISUAL DISTURBANCES ON CMU-MOSEI DATASET

| <i>Setting</i>   | <i>MAE</i> | <i>Corr</i> | <i>F1</i>   |
|------------------|------------|-------------|-------------|
| <b>RAF-Net</b>   | 0.518      | 0.804       | 83.80/87.39 |
| w/o Vision       | 0.552      | 0.768       | 80.50/84.85 |
| Vision (Noise)   | 0.795      | 0.750       | 80.15/83.90 |
| Vision (Shuffle) | 0.520      | 0.801       | 83.65/87.25 |

#### G. Hyperparameter Sensitivity Analysis

We analyze the sensitivity of the hyperparameter  $\epsilon$  to evaluate the stability of the model under different sparse constraints. As shown in Table 5. when  $\epsilon = 1e-6$ , the model achieves the best performance on all indicators, indicating that the best balance has been achieved between noise filtering and key emotional clue retention. Deviating from this value will lead to a decrease in performance: higher  $\epsilon$  will over-constrain

cross-modal correlation, while lower  $\epsilon$  will lead to overfitting by introducing redundant, near-zero interactions.

TABLE V. RESULTS OF SENSITIVITY ANALYSIS OF  $\epsilon$  ON THE CMU-MOSI DATASET

| $\epsilon$ | MAE   | Corr  | ACC-2       | F1          |
|------------|-------|-------|-------------|-------------|
| 1e-8       | 0.709 | 0.782 | 81.30/84.10 | 81.25/83.95 |
| 1e-6       | 0.698 | 0.843 | 86.44/88.11 | 86.45/88.16 |
| 1e-4       | 0.730 | 0.815 | 84.10/86.20 | 83.95/85.85 |
| 1e-2       | 0.748 | 0.776 | 82.10/84.20 | 81.90/84.10 |

#### H. Visualizing representations

To qualitatively evaluate the characterization quality, we will visualize the t-SNE projection of the final fusion features on the test set. As shown in the Fig. 2, RAF-Net has built a well-structured embedded space, in which negative samples (blue) and non-negative samples (red) clustering have obvious distinctions. It is worth noting that even in a noise-intensive CMU-MOSEI environment, the model maintains clear decision-making boundaries and minimizes category entanglement. This confirms that RAF-Net learns from a robust, noise-resistant manifold rather than simply remembering discrete labels.

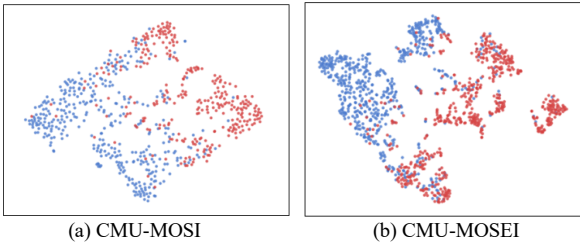


Fig. 2. t-SNE visualization on (a) CMU-MOSI and (b) CMU-MOSEI test sets.

#### V. CONCLUSION

This paper proposes a multi-modal Sentiment analysis framework RAF-Net, which effectively solves the problem of cross-modal reliability difference between text and acoustic flow by breaking the traditional symmetric fusion paradigm and setting visual clues as global reliability anchor points. Experiments prove that the framework has excellent robustness. Although setting vision as a static anchor point greatly reduces computational redundancy, there are still limitations in capturing instantaneous emotional features such as irony and metaphors. In the future, a lightweight dynamic perception module will be introduced to realize the evolution from static anchor points to dynamic perception.

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